

NAVEN, Sweden

WMO: 024030

Lat: 58.700N

Lon: 13.111E

Elev: 53

StdP: 100.69

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-10.8	-8.8	-13.1	1.2	-9.9	-11.2	1.4	-8.2	12.4	1.2	11.3	1.6	2.8	70	0.618

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.9	25.2	18.9	23.5	18.2	21.9	17.2	19.8	23.6	18.9	22.3	18.1	21.0	2.8	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.5	21.5	17.6	12.7	20.8	16.7	12.0	20.0	57.1	23.5	54.0	22.3	51.2	21.2	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.3	10.0	8.8	-14.1	27.8	5.1	1.5	-17.8	28.8	-20.7	29.7	-23.6	30.5	-27.3	31.6		
(5)			-13.9	21.0	4.6	1.2	-17.2	21.9	-19.9	22.6	-22.5	23.3	-25.9	24.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.9	-0.8	-1.1	1.5	5.7	11.0	15.4	18.0	17.1	13.4	8.3	4.1	0.9		
	DBStd	7.41	3.67	4.05	3.33	2.63	2.97	2.55	2.52	2.23	2.37	2.90	2.92	3.87		
	HDD10.0	1590	333	310	264	130	22	0	0	0	3	68	177	283		
	HDD18.3	3873	592	543	523	378	227	94	38	51	147	312	427	541		
	CDD10.0	807	0	0	0	2	53	163	247	221	106	15	1	0		
	CDD18.3	48	0	0	0	0	1	7	27	13	1	0	0	0		
	CDH23.3	161	0	0	0	0	3	30	96	31	1	0	0	0		
	CDH26.7	14	0	0	0	0	0	4	9	2	0	0	0	0		
(14) Wind	WSAvg	4.1	4.4	4.2	4.1	3.8	3.7	3.7	3.7	3.7	4.2	4.4	4.7	4.6		
(15) Precipitation	PrecAvg	650	46	40	31	41	46	70	75	70	61	64	55	51		
	PrecMax	814	82	80	69	92	88	159	182	137	149	138	95	112		
	PrecMin	499	7	8	5	7	9	6	7	18	10	10	10	15		
	PrecStd	81	22	17	14	27	25	33	42	34	37	31	23	24		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.5	7.1	10.7	16.2	23.0	26.6	27.8	25.9	21.8	16.1	11.6	8.5		
		MCWB	5.8	6.1	6.8	10.0	15.2	18.2	20.1	19.2	16.7	13.2	10.8	7.9		
	2%	DB	5.4	5.7	8.2	12.9	20.1	23.5	25.9	24.0	19.7	14.0	10.0	7.3		
		MCWB	4.8	5.0	5.5	8.9	14.2	17.4	19.5	18.9	16.0	12.5	9.0	6.6		
	5%	DB	4.5	4.5	6.7	11.1	17.7	21.3	24.1	22.4	18.2	13.1	9.1	6.4		
		MCWB	3.9	3.7	4.9	7.7	13.2	15.9	18.7	17.9	15.2	11.9	8.3	5.7		
	10%	DB	3.8	3.6	5.5	9.7	15.7	19.7	22.3	21.1	17.0	12.2	8.2	5.5		
		MCWB	3.2	2.9	4.1	6.9	12.0	14.9	17.9	17.3	14.5	11.1	7.5	4.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.1	6.2	7.2	11.0	16.5	19.4	21.4	20.7	17.9	14.2	11.0	8.2		
		MCDB	6.3	6.6	9.2	14.3	21.4	24.8	25.8	23.8	20.6	15.4	11.4	8.4		
	2%	WB	5.0	5.1	6.1	9.3	14.9	17.9	20.3	19.5	16.7	12.9	9.2	6.9		
		MCDB	5.2	5.7	7.6	12.2	18.9	22.5	24.2	22.6	18.8	13.7	9.7	7.3		
	5%	WB	4.0	3.9	5.1	8.2	13.7	16.7	19.4	18.6	15.6	12.1	8.5	5.8		
		MCDB	4.4	4.3	6.5	10.4	17.1	20.4	23.0	21.3	17.7	12.9	8.9	6.2		
	10%	WB	3.2	3.1	4.2	7.2	12.6	15.7	18.5	17.8	14.7	11.2	7.6	5.0		
		MCDB	3.7	3.7	5.2	9.2	15.2	18.6	21.7	20.4	16.6	12.0	8.1	5.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.2	3.6	5.1	6.3	7.0	7.4	6.9	6.4	5.3	3.9	3.0	3.1		
		MCDBR	3.2	3.5	6.6	8.3	9.8	9.9	9.4	8.3	6.9	4.6	3.6	3.3		
	5% WB	MCWBR	3.3	2.9	4.4	5.0	5.4	5.2	4.3	4.3	4.1	3.4	3.7	3.5		
		MCWBR	3.3	3.2	6.1	7.4	8.6	9.0	8.3	7.2	6.6	4.2	3.5	3.3		
(40) Clear-Sky Solar Irradiance	taub		0.266	0.303	0.326	0.366	0.362	0.353	0.373	0.368	0.339	0.326	0.291	0.249		
		taud	2.225	2.308	2.401	2.361	2.429	2.479	2.438	2.451	2.516	2.474	2.318	2.155		
	Ebn at Noon		566	725	822	841	870	884	854	832	804	699	533	448		
		Edh at Noon	45	69	84	103	103	99	102	93	75	59	42	34		
(44) All-Sky Solar Radiation	RadAvg		0.33	0.90	2.43	3.90	5.13	5.81	5.18	4.08	2.76	1.24	0.44	0.23		
	RadStd		0.06	0.16	0.24	0.49	0.45	0.31	0.65	0.37	0.33	0.20	0.09	0.02		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page